System Stewardship

COMPUTER SYSTEMS LITERACY MODULE 5

Why?

Computer users need to learn system stewardship because modern computing systems are shared, complex, and long-lived, where small technical decisions can have large consequences for security, reliability, cost, and ethics. **Stewardship** means taking responsibility for how systems are designed, used, and maintained over time—thinking beyond individual tasks to system-wide impact, long-term sustainability, and the people who depend on those systems.

1. **Security & Reliability**: Most system failures stem from human misconfiguration, not hardware. Stewardship reduces these risks.
2. **Shared impact:** Actions by one user or developer can affect many others, so systems must be treated as shared resources, not personal tools.
3. **Risk prevention:** Minor mistakes can trigger major failures; stewardship encourages careful, defensive decision-making.
4. **Long-term sustainability:** Systems should remain maintainable, understandable, and scalable well into the future.
5. **Ethical responsibility:** Computing systems handle sensitive data and critical services, requiring responsible and ethical management.

Agenda

1. Software Hygiene Practices
2. Update and Patch Discipline
3. Storage & File System Care
4. Startup & Performance Management
5. Security Basics
6. Backup & Recovery Thinking
7. Hardware Awareness
8. Digital Discipline

The Art of Software Hygiene

Goal: Learn how to invite the right guests and show them the door properly when they leave.

- Software is not just a file. It is a guest. When you install a program, you are giving it permission to change your computer's "state".
- A **System State** is the snapshot of a system at any given moment. It is the collective sum of all the settings, files, and background processes that define what the computer is currently doing and how it is configured.

The System State

What happens when you click "Install"?

Software doesn't just sit in one folder. It weaves itself into the system:

1. **The Disk:** Files are copied to Program Files, AppData, and Temporary folders.
2. **The Registry/Config:** The "brain" of Windows/OS gets new instructions on how the app should behave.
3. **Startup:** Some apps "claim a seat" and start running the moment you turn on your computer, even if you don't open them.

System State: Registry

Think of your computer not just as a box of files, but a complex machine that needs a **rulebook** to function. On Windows, this rulebook is the registry. On other systems like MacOS or Linux, these rules are often stored in **Configuration (config) Files**.

- In Windows, the "registry" is a massive centralized database of settings. Instead of every app keeping its own separate notes, Windows uses the Registry as a centralized master database.
- It is organized like a massive tree of folders (called "keys") and files (called "values")
- When you install an application, it adds new rules to this database. E.g., "Always open .pdf files with this app."

Bloatware – The Uninvited Guests

"Just because it came with the computer, does not mean it belongs there."

What is it?: Software pre-installed by the manufacturer (OEM) or carrier that you didn't ask for (e.g., trial antivirus, branded "assistants," or random games).

The Performance Tax

- Resource Drain: Many bloatware apps launch automatically at startup, eating up your RAM and CPU, before you even open a browser.
- Storage Clutter: These apps take space in your file system.
- Security Holes: Pre-installed software is rarely updated by the user, creating unpatched vulnerabilities in your fortress.

How to Remove Them: Sort apps by manufacturer, and research and uninstall if you don't recognize any application. Disable anything that does not need to be running 24x7 using the Startup tab in the Task Manager. Pro Tip: Fresh Start (installing a fresh copy of the OS).

Two Ways to Run Software

Installers vs. Portable Apps

Feature	Installers (.exe, .msi, .dmg)	Portable Apps (.zip, .exe)
Integration	Deep. Changes Registry and system settings.	Shallow. Stays inside its own folder.
Experience	Best for main tools (Browsers, Office).	Best for utilities on a USB drive.
Removal	Requires an "Uninstaller" to clean up.	Just delete the folder and it’s gone.
Updates	Usually automatic or via system tools.	Usually manual.

The Proper Goodbye

Why you can't just "Delete" a folder

If you simply delete a program's folder in Program Files:

Digital "Gunk": The Registry entries remain, looking for files that aren't there.

Ghost Processes: Startup items might still try to run, causing errors.

Wasted Space: Hidden configuration files in AppData stay forever.

The Rule: Always use the **official Uninstaller** or the **System Settings (Add/Remove Programs)**. The uninstaller is a script that "undoes" the installation steps.

The "Trusted Source" Test

Knowing Which Apps Are Safe

Before clicking download, ask:

Is this the Official Site? (Avoid "Download Hubs" that add their own wrappers).

Is it "Bundled"? Many free installers include "Recommended" toolbars or cleaners.

The Crack Trap: "Cracked" or "Pirated" software often includes malware that changes your system state in ways you can't see (like turning your PC into a botnet).

Identifying Genuine Software

When a major developer (like Microsoft, Google, etc.) has not created a tool you need, you need to act like a "Digital Detective" and collect technical evidence.

1) The "crowdsourced" scan (Virus Total): Before running the file, upload the installer or the .exe file to VirusTotal.com

2) Check the Digital Signature: Right click the file and select Properties. Look for a tab called Digital Signatures. You want to see a name in the "Name of the Signer" field.
   1. The Digital Signature proves that the code hasn't been tampered with since the developer finished it.
   2. If the tab is missing or says "Unknown Publisher" when you run it, the developer hasn't verified their identity. This doesn't mean that it's a virus, but it means you are taking a much higher risk.

Identifying Genuine Software

3. The "Community Footprint" Research: Go to Google or Reddit and search "[App Name]" + Safe or "[App Name]" + review.

- Stars and Reviews: If the app has a high rating, it usually means people trust it. Look at the "issues" tab – are people reporting malware, or just regular bugs?

4. The Sandbox Detonation (Advanced): If you are still unsure and really need the app, do not run it on your main "System State". Use a Sandbox.

- Windows Sandbox: A built-in feature of Windows Pro/Enterprise that creates a "disposable" computer inside your computer. Run the app there and monitor the events.
- Any.run: A website that lets you "watch" a file run in a virtual cloud computer to see exactly what files it changes and what websites it tries to connect.

The Hidden Cost of "Free"

If you aren't paying, what are you trading?

Software takes time and money to build. If it's free, you are often paying with:

Your Control: The software might change your default browser or search engine.

Your Privacy: It may "phone home" with data about your usage.

Your Resources: It might run background processes that slow down your PC to show you ads.

Pro Tip: Choose **Open-Source** software (like VLC or LibreOffice) when possible—it's free because of community, not because of data tracking.

Summary Checklist

Good Hygiene Habits

[ ] **Check the source:** Download only from official developers.

[ ] **Read the prompts:** Opt-out of "extra" bundled software during installation.

[ ] **Go Portable:** Use portable or online versions for one-time tasks.

[ ] **Uninstall Cleanly:** Never just delete a folder; use the uninstaller.

[ ] **Audit Startup:** Periodically check your Task Manager to see what's running when you boot up.

Updates & Patch Discipline

Concepts

Vulnerabilities are discovered *after* software ships

Updates reduce attack surface

Delaying updates = accumulating risk

Updates & Patch Discipline

Updates are a form of maintenance, not decoration.

1. Checking OS Update Status: Operating Systems handle the most critical permissions (drivers, kernel, network stack). An unpatched OS is an open door for privilege escalation attacks.
2. Updating browsers and critical apps.
3. Understanding automatic vs manual updates.

Why We Patch

Slide 1: Why We Patch (The Reality of Software)

Software is never "finished."

Vulnerabilities are discovered *after* shipping: No code is perfect. Security holes (bugs) are often found months or years after a program is released.

The "Race" Begins: Once a vulnerability is made public, hackers race to exploit it, while developers race to "patch" (fix) it.

Updates = The Cure: A patch is a piece of code designed to plug a security hole. Without it, your system remains "sick" and exposed.

Reducing Your Attack Surface

Delaying updates = Accumulating risk.

What is an Attack Surface? This is the total sum of all points where an unauthorized user can enter your system.

Shrinking the Target: Every update you install closes a door. The more updates you skip, the larger your "attack surface" becomes.

The Risk Compounder: As software ages without patches, the "blueprints" for how to hack it become common knowledge. Delaying a 10-minute update for a month gives attackers a 30-day window to find you.

Practical Skills

Taking control of the maintenance.

OS Update Status: Your Operating System (Windows/macOS) is your first line of defense.

- *Skill:* Know how to manually "Check for Updates" to ensure your PC hasn't paused them.

The Browser Barrier: Your web browser is your most-used tool and your most-attacked. Ensure Chrome, Firefox, or Edge are set to update automatically.

Auto vs. Manual:

- **Automatic:** Best for security. The system stays safe without you thinking about it.
- **Manual:** Use only for specialized "critical" apps where a new version might break your specific work setup.

Maintenance, Not Decoration

Shifting your mindset.

It’s Not About Features: Many users ignore updates because they don’t see "new buttons" or "prettier icons."

Maintenance is Invisible: Think of updates like an oil change for a car. You don't see it working, but it prevents a total engine failure.

The Golden Rule: If a prompt asks to update a security-sensitive app (Bank, Email, OS), **do it today.**

The Final Step: Remember that most updates require a **Restart** to actually take effect. Your system isn't safe until the reboot is finished.

Storage and File System Care

Core skills

1. Checking free disk space
2. Identifying large files and unused apps
3. Using built-in cleanup tools
4. Understanding user folders vs system folders

Concepts

1. Filesystems degrade in usability when full
2. OS needs free space for temp files, swap, updates
3. Deleting the wrong folder can break software

The Logistics of Disk Space

Why "full" means "slow"

Disks fill up logically, not magically: Every file, cache, and update occupies physical sectors on your drive.

The 10-15% Rule: An Operating System (OS) requires "breathing room" for:

- **Swap/Virtual Memory:** Using disk space as temporary RAM when physical RAM is full.
- **Temporary Files:** Files created during app installation or video rendering.
- **Update Staging:** Space to download and unpack system patches.

The Consequence: When free space drops below ~10%, the OS struggles to manage these background tasks, leading to system hangs and "Disk Full" errors.

Navigating the File System

User Folders vs System Folders

User Folders (`C:\Users\[Name]`): This is your "Safe Zone." It includes Documents, Downloads, Desktop, and Pictures. Deleting files here only affects YOUR data.

System Folders (`C:\Windows` and `C:\Program Files`): This is the "Engine Room."

Contains the code that runs the OS and your apps.

The Danger: Deleting folders here manually instead of using an uninstaller can break software dependencies or prevent the OS from booting.

The AppData Hidden Folder: Located inside your User folder, it stores your specific settings. Cleaning this requires specialized tools to avoid losing saved progress or passwords.

Identifying & Cleaning Clutter

Check Free Space: Use "This PC" (Windows) or "About This Mac" to see a high-level view of storage distribution.

Targeting Large Files: Use built-in tools like **Storage Settings** to see a breakdown of what is taking up space (System, Apps, or Large Files).

Built-in Cleanup Tools:

- **Windows Disk Cleanup / Storage Sense:** Safely removes system logs, temporary internet files, and emptied Recycle Bin items.
- **The Benefit:** These tools know which "System" files are safe to delete and which must stay.

Unused Apps: Periodically audit your installed apps list. An app you don't use is just "State Bloat" taking up valuable space.

Fragmentation & File Health

Maintaining Data Order

What is Fragmentation? Over time, files are scattered across different physical locations on a drive rather than being stored in one continuous block.

SSD vs. HDD:

HDD (Mechanical): Needs "Defragmentation" to physically move pieces of files closer together for the reading head to find them faster.

SSD (Flash): Does **not** need defragmenting. Instead, it uses **TRIM** (Optimization) to tell the drive which data blocks are no longer in use.

The Habit: Modern OSs schedule this automatically, but you should verify that "Optimization" is turned on and running weekly.

Startup & Performance Management

Slowness is often self-inflicted.

Core skills

1. Viewing startup programs
2. Disabling non-essential startup apps
3. Recognizing background vs foreground processes
4. Checking CPU, RAM, disk usage at a basic level

Concepts

1. Limited resources are shared
2. Boot time = sum of startup tasks
3. Performance ≠ CPU speed alone

The Performance Equation

Why "Slowness" is often self-inflicted

Shared Resources: Your CPU, RAM, and Disk have a fixed capacity. Every open app and background task "takes a slice" of that pie.

Performance ≠ CPU Speed Alone: A fast processor cannot fix a "resource logjam" caused by too many apps competing for Disk access or RAM.

The Resource Hierarchy:

- **CPU:** The "Speed" (How fast tasks are processed).
- **RAM:** The "Workspace" (How many tasks can stay open at once).
- **Disk:** The "Retrieval" (How fast data is loaded into the workspace).

Managing the Boot Process

Boot Time = The sum of startup tasks

The Startup Chain: Every app marked as "Enabled" at startup adds seconds to your boot time. If 20 apps start with Windows, your computer isn't "ready" until the last one finishes loading.

Core Skill: Viewing Startup Apps:

- **Windows:** Task Manager (Ctrl+Shift+Esc) > Startup tab.
- **macOS:** System Settings > General > Login Items.

The "Impact" Rating: Look for "High Impact" apps. These are the primary culprits for a slow start.

Action: Disable anything that isn't essential. (You can still open the app manually when you actually need it).

Background vs. Foreground

Understanding what is currently "Active"

Foreground Processes: The apps you are actively using (e.g., your Browser, a Word Doc). You want the most resources directed here.

Background Processes: Tasks that run "behind the scenes."

- *Good:* Security scans, system updates, cloud syncing.
- *Bad:* Bloatware, unneeded updaters, and "zombie" processes from apps you closed but didn't actually exit.

The Audit: Periodically check the **Processes** tab in Task Manager. If an app you aren't using is using >5% of your CPU, it is a candidate for being disabled or uninstalled.

Resource Monitoring

Recognizing the Bottleneck:

100% CPU: Too many heavy calculations (e.g., video export, malware).

100% RAM: You have too many tabs or apps open; the system is swapping to the disk.

100% Disk: Often caused by system updates or a failing drive; makes the whole OS feel "frozen."

The Rule of Thumb: If your computer feels slow, check the **Performance Tab** before restarting. Identifying *which* resource is maxed out tells you *which* software habit to change.

Security Basics

Core skills

1. Using built-in antivirus
2. Recognizing phishing and fake installers
3. Understanding permissions (admin vs standard user)
4. Safe browsing habits

Concepts

1. Most infections are social engineering
2. Antivirus is reactive, not omniscient
3. The user is part of the security boundary

Security is mostly behavior.

The Human Firewall

The User as a Boundary: You are the final gatekeeper. No security software can fully protect a system if the user manually bypasses warnings or grants malicious permissions.

Social Engineering: Most modern infections do not "hack" the computer; they "hack" the human. They use fear (fake virus alerts), urgency (account locked), or greed (free software) to trick you into lowering your guard.

Reactive vs. Omniscient: Antivirus software works primarily by recognizing *known* threats. It is "reactive." New, undiscovered "Zero-Day" threats can only be stopped by cautious user behavior.

Built-in Defense & Permissions

Protecting the System "Brain"

Built-in Antivirus (Windows Defender): Modern built-in tools are highly effective and less resource-heavy than third-party suites. Ensure "Real-time protection" is always toggled ON.

Admin vs. Standard User:

- **Standard User:** Can run apps and save files but cannot change system-wide "state."
- **Administrator:** Has the power to modify the Registry, system folders, and security settings.

The "Least Privilege" Rule: Daily work should be done in a Standard account. Only provide Administrator credentials (UAC prompt) when you are 100% certain why the app needs to change the system.

Spotting the Trap

Recognizing Phishing and Fake Installers

Phishing: Emails or sites designed to steal credentials. Check the **sender's actual email address** (not just the name) and hover over links to see the true destination URL.

The "Fake Installer" Strategy: * Avoid "Free Download" buttons on unofficial sites. These often lead to "Wrappers" that install the app you want *plus* three pieces of bloatware or malware.

- **Cloned Sites:** Attackers create perfect copies of official websites (like "https://www.google.com/search?q=vlc-download.com" instead of "videolan.org") to distribute infected installers.

Check the URL: Always verify the domain name in the address bar before entering data or downloading.

Safe Browsing Habits

Reducing the Attack Surface

HTTPS is the Minimum: Look for the padlock icon. It ensures your connection is encrypted, but remember: *A malicious site can still have a padlock*.

Extension Hygiene: Browser extensions have deep access to your web data. Only install extensions from trusted developers and remove any you don't use.

Ad-Blockers as Security: Many "malvertising" attacks occur through infected ads on legitimate sites. Using a reputable ad-blocker removes this entire category of risk.

The "Think Before You Click" Protocol: If a site demands you disable your antivirus or "Run as Administrator" to view content, it is almost certainly malicious.

Backup & Recovery Thinking

Core skills

1. Backing up important files
2. Knowing where personal data actually lives
3. Restoring files from backup
4. Understanding cloud sync vs backup

Concepts

1. Hardware *will* fail
2. Deletion ≠ immediate destruction
3. Backups are time machines, not mirrors

Mistakes are inevitable; loss shouldn't be.

The Physics of Failure

Hardware will fail: Every storage drive (SSD or HDD) has a finite lifespan. It is not a question of *if* it will fail, but *when*.

The "Time Machine" Concept: A true backup is a "Point-in-Time" snapshot. If you accidentally delete a file today, you need a way to go back to "yesterday" to retrieve it.

Deletion ≠ Immediate Destruction: When you delete a file, the OS usually just marks that space as "available." The data often still exists until new data is written over it, but you cannot rely on this for recovery.

Locating your Data

Knowing where your personal data lives.

The User Profile: Most personal data is concentrated in `C:\Users\[YourName]`. This includes:

Documents, Pictures, Desktop, and Downloads.

Hidden Data (AppData): Many apps store critical data—like browser bookmarks, game saves, and email archives—in the hidden `AppData` folder.

The "External" Trap: Remember that data saved on USB sticks or SD cards is often *only* in that one location and is not covered by standard system backups.

Core Skill: Practice identifying which folders on your drive actually contain "irreplaceable" data versus "re-installable" software.

Cloud Sync vs. True Backup

Backups are snapshots.

Cloud Sync (OneDrive, Dropbox, iCloud):

How it works: It "mirrors" your files. If you delete a file on your laptop, it is immediately deleted from the cloud.

Risk: Syncing is great for access, but dangerous for accidental deletion or "ransomware" that encrypts your files.

True Backup:

How it works: It creates a separate, read-only copy of your files at a specific time.

Benefit: If you catch a virus or delete a folder, the backup remains untouched and ready for "Restoration."

The 3-2-1 Rule: 3 copies of your data, on 2 different media types, with 1 copy stored off-site (cloud or different building).

Recovery Readiness

The "Restoration" Skill

Backups are useless if they don't work: A backup is only as good as your last successful "Restore" test.

Core Skill: Restoring Files:

- Learn how to navigate your backup software to pick a single file from three days ago and bring it back to your desktop.

Version History: Many modern systems (Windows File History / macOS Time Machine) allow you to right-click a file and select "Previous Versions." This is your first line of defense for accidental edits.

The Habit: Set your backup to run automatically. Manual backups are rarely performed often enough to be reliable.

Hardware Awareness

Enough to avoid damage and diagnose basics.

Core skills

1. Safe shutdown vs force shutdown
2. Proper charging and battery care
3. Connecting/removing peripherals safely
4. Basic troubleshooting (is it software or hardware?)

Concepts

- Hardware operates within physical limits
- Power loss affects data integrity
- Heat and dust shorten lifespan

Physical Limits & Longevity

Heat and Dust: The Silent Killers

Thermal Management: Hardware generates heat as it processes data. If that heat isn't dissipated, the system "throttles" (slows down) to prevent physical melting.

Airflow: Dust acts as insulation, trapping heat inside components.

- **The Habit:** Keep laptops on hard, flat surfaces. Using a laptop on a bed or carpet blocks intake vents and can cause permanent component degradation over time.

Power and Data Integrity: * **Safe Shutdown:** Allows the OS to "park" the drive heads and finish writing to the **Registry**.

- **Force Shutdown (Holding Power):** Cuts power instantly. If the "brain" was in the middle of writing a rule, that file becomes corrupted, leading to system errors.

Battery Care & Charging

Managing Chemical Lifespans

Cycle Counts: Every battery has a limited number of "cycles" (0% to 100% charges) before it loses the ability to hold a charge.

The 20-80 Rule: Lithium-ion batteries are most stressed at the extremes (0% or 100%). Keeping your battery between 20% and 80% can significantly extend its total lifespan.

Heat is the Enemy: Charging your laptop in a hot environment or under heavy load (like gaming) accelerates battery chemical breakdown.

Deep Discharge: Never leave a device at 0% for weeks; the battery can "sleep" permanently and refuse to take a charge again.

Peripherals & Ports

Connecting and Removing Safely

Static Discharge: Touching a port after walking on carpet can send thousands of volts into the motherboard.

Safe Removal:

- **Why:** The "Safely Remove Hardware" icon tells the OS to stop writing data to a USB drive. Pulling a drive while the "write" light is blinking is the #1 cause of corrupted files.

Mechanical Stress: USB and charging ports are soldered to the motherboard. Repeatedly "yanking" cables or using them at sharp angles can break the solder joints, requiring expensive repairs.

Basic Troubleshooting Logic

Is it Software or Hardware?

The "Reboot" Test: If a problem disappears after a restart, it was likely a **Software** issue (a temporary "state" error).

Software Indicators: * Errors only happen in one specific app.

- The system is slow, but the fans aren't spinning loudly.

Hardware Indicators: * The computer shuts off suddenly without an error message (Overheating/Power).

- Strange noises (grinding fans or clicking HDDs).
- Visual artifacts (lines or "snow" on the screen).

External vs. Internal: Always unplug all peripherals (mouse, printer, USB) to see if the problem persists. Often, a "broken" computer is actually just a "broken" USB cable.

Digital Discipline & Judgement

This is the hidden curriculum.

Core skills

1. Reading dialogs before clicking “Next”
2. Distinguishing system messages from ads
3. Knowing when to stop and ask for help

Concepts

- Computers do exactly what they’re told
- Carelessness compounds over time
- Literacy ≠ memorization of steps

The Myth of Computer "Mistakes"

Computers do exactly what they are told

Logic, Not Intent: A computer does not know what you *meant* to do; it only executes the specific command you gave it.

The Compounding Effect: Digital carelessness (skipping updates, ignoring errors, messy file naming) compounds over time. Small, messy habits eventually lead to a system that is slow, unstable, or compromised.

Literacy vs. Memorization: True digital literacy is not about memorizing which buttons to click in order. It is about understanding the **logic** of the interface so you can navigate software you have never seen before.

The Next-Next Finish Trap

Reading Dialogs and Checkboxes

The Speed Tax: Most "bundled junk" and unwanted system changes happen because the user clicked "Next" too quickly.

The "Custom" Secret: Always choose "Custom" or "Advanced" installation. This reveals hidden checkboxes for toolbars, search engine changes, and data-tracking "opt-ins."

Active Reading: If a dialog box appears, identify three things before clicking:

- **The Source:** Who is asking? (The OS or a website?)
- **The Request:** What does it want to change? (Files? Permissions? Startup?)
- **The Risk:** What happens if I say "No"? (Most of the time, the app will still work fine).

Signal vs. Noise

Distinguishing System Messages from Ads

The Visual Language of Authority: * **System Messages:** Usually follow the OS design language (consistent fonts/colors). They rarely use "flashing" text or high-pressure language.

- **Ads/Scareware:** Often appear in the browser window, use bright red "Warning" colors, and claim you have "37 viruses found."

Context is Key: If you are browsing a website and a "System Alert" appears *inside* the webpage, it is an advertisement.

Core Skill: Learn to recognize the specific "User Account Control" (UAC) and "System Notification" styles of your specific Operating System.

Knowing When To Stop

The "Ask for Help" Threshold

The Point of No Return: Some actions (like formatting a partition, editing the Registry, or "Flashing" hardware BIOS) can permanently break a system if done incorrectly.

The Search Skill: If you see an error code (e.g., `0x800f081f`), do not guess. Copy the code into a search engine to find the documented solution before trying "random" fixes.

The Threshold: If you find yourself following a tutorial that asks you to disable your Antivirus or Firewall, **stop.** This is the moment to ask a professional or a more experienced peer for a second opinion.

Minimal Graduation Standard

A student should be able to:

1. Install and remove software safely
2. Keep a system updated
3. Free space and improve startup time
4. Avoid obvious security traps
5. Recover from simple mistakes

Information & Action Items

- Create a LinkedIn account and send me a connect request so that I can add you to the Project Mercy LinkedIn Group.
- Assignments are due tonight at 11:59 PM. Deliverables:
  - Module 1 Lab Work
  - Assignments on Binary and Hexadecimal Conversion.
- Subscribe to technology newsletters.